

Sr. No.	Topic	Objective of the Session	Focused Technical Skill	Follow-up Assignment / Practice Task(Company Tags)		
				EASY	MEDIUM	HARD
1	Overview of Databases: DBMS, Architecture (Three-Level ANSI-SPARC), Data Independence	Applying 3-schema architecture, ANSI-SPARC, Data Independence	Logical & Physical Data Independence, 2-tier & 3-tier architecture	https://platform.stratascratch.com/coding/10552-identify-never-sold-inventory-products?code_type=1 (Google)	https://platform.stratascratch.com/coding/10365-common-friends-script?code_type=1 (Google)	https://platform.stratascratch.com/coding/2165-sales-percentage-weeks-beginning-and-end?code_type=1 (Meta)
2	DBA Responsibilities	Roles and responsibilities of a Database Administrator	DBA role and data management	https://platform.stratascratch.com/coding/10549-top-customers-with-dense-ranking?code_type=1 20(IBM)	https://datalemur.com/questions/rolling-average-tweets (Twitter)	https://datalemur.com/questions/repeated-payments (Stripe)
3	Relational Data Structure: Relations, Attributes, Keys, Schema, Instances, Integrity Constraints	Apply relational data structures and integrity constraints	Key Constraints, RDBMS vs DBMS, KEYS, Foreign Key Constraints, Referential Integrity, Cascading Operations	https://www.codechef.com/learn/course/sql-intermediate/SQ00BS02/problems/SQLKEY01C?tab=statement	https://www.codechef.com/learn/course/sql-intermediate/SQ00BS02/problems/GSQ64	https://www.codechef.com/learn/course/sql-intermediate/SQ00BS03/problems/GSQ68?tab=statement

4	KARGIL COMPANY SPECIFIC EXPERIMENT 1 [SQL COMMANDS]	Implement DDL, DML, and DCL queries with Integrity Constraints	DDL, DML, DQL & DCL SQL Queries.			
5	Data Models: Relational, Hierarchical, and Network Models	Compare and contrast relational, hierarchical, and network models	Data model structures			
6	ER Model: Design, Mapping Constraints, ER Diagrams	ER modeling concepts, including generalization and aggregation	ER diagram creation & Mapping constraints			
7	KARGIL COMPANY SPECIFIC EXPERIMENT 2 [Set Operations]	Implement SQL set operations (UNION and UNION ALL) to combine result sets and manage duplicate records effectively	Union, Union All ,			
8		Implement INTERSECT and EXCEPT operations to identify common records and differences between multiple datasets.	Intersection ,Except			
9	KARGIL COMPANY SPECIFIC EXPERIMENT 3 [GROUPING , ORDERING and Aggregate Function with Conditional Statement]	Implement SQL SELECT queries using WHERE, GROUP BY, HAVING, ORDER BY, and Aggregate Functions with Conditional Statements to filter, group, analyze, and sort data effectively.	GROUP BY + HAVING, ORDER BY, WHERE CLAUSE, and Aggregate Function with Conditional Statement			
10		Implement filtering of records using conditional logic and handle duplicate data.	SQL WHERE clause, DISTINCT, ORDER BY,			
11	Relational Algebra: Join Operations	Join operations, including inner, outer, and cross joins	CROSS JOIN, CROSS PRODUCT, NATURAL JOIN, LEFT & RIGHT OUTER			

12	Relational Calculus: Tuple and Domain Calculus	Tuple and domain calculus concepts	Relational calculus			
13	KARGIL COMPANY SPECIFIC EXPERIMENT 4 [Join Operations]	Implement INNER JOIN and LEFT JOIN operations to combine data from related tables and retrieve matching as well as non-matching records.	INNER JOIN ,LEFT JOIN			
14		Implement RIGHT JOIN and FULL OUTER JOIN operations to combine data from multiple related tables and retrieve matching as well as non- matching records	RIGHT JOIN , FULL OUTER ,JOIN			
15		Implement CROSS JOIN and SELF JOIN operations to combine data from related tables and retrieve meaningful business information.	CROSS JOIN ,SELF JOIN			
16		Calculate the percentage contribution of a cuisine to total revenue	Arithmetic Calculations in SQL			
17	KARGIL COMPANY SPECIFIC EXPERIMENT 5 [Arithmetic Calculation, PL-SQL , STRING MANIPULATION]	Develop PL/SQL programs with block structure, variables, and exception handling	PL/SQL programming: DECLARE, BEGIN & END blocks			
18		Implement SQL string functions (LENGTH and CHAR_LENGTH) along with the WHERE clause to validate text data based on string length constraints.	SQL string functions (LENGTH, CHAR_LENGTH), WHERE clause			
19	Functional Dependencies: Trivial, Non-Trivial, Transitive	Understand functional dependencies and their impact on schema design	Schema dependency analysis			

20	Normalization: 1NF, 2NF, 3NF	Learn normalization techniques to optimize schema design	Normalization process			
21	Normalization: BCNF, Multi-Valued and Join Dependencies	Explore advanced normalization forms (BCNF, 4NF, 5NF)	Advanced normalization			
22	PL-SQL CONDITIONAL STATEMENTS	Implement control structures in PL/SQL (If-Else, Case)	IF-ELSE, ELSE-IF, ELSE-IF LADDER, CASE STATEMENTS in PL-SQL BLOCK			
23	Normalization: Lossy vs Lossless Decomposition	Understand the difference between lossy and lossless decomposition	Schema decomposition			
24	Database Security: Introduction and Threats	Security challenges in databases	Understanding database security			
25	Quiz 1	Quiz on SQL Introductory Part	Assessment			
26	[PL-SQL ITERATIVE STATEMENTS]	Implement PL/SQL loops (For, While, Do-While)	PL/SQL iterative programming			
27		How to analyze numerical data using conditional logic in SQL	Modulus operator (MOD / %)			
28	Control Structures: Conditional and Iterative	Conditional and iterative control in SQL	FOR, DO, WHILE with CASE & IF-ELSE			
29	Control Structures: Sequential Control	Sequential control flow concepts	Sequential programming logic			
30	Cursors: Declaring, Opening, Fetching, Closing	Understand the concept and application of cursors	Cursor-based programming			
31	KARGIL COMPANY SPECIFIC EXPERIMENT 6 [PL-SQL CURSORS]	Implement cursors in PL/SQL	Cursor programming			
32		Implement implicit and explicit cursors along with cursor attributes to perform row-by-row data processing in PL/SQL	Implicit Cursors, Explicit Cursors, Cursor Attributes			

33	Views: Creation and Manipulation	Concept and use of views in databases	View creation and management			
34	KARGIL COMPANY SPECIFIC EXPERIMENT 7 [VIEWS IN SQL]	Implement views in SQL	SQL view creation and query execution			
35		Design a materialized view, compare and analyze the execution time with simple, complex and materialized view	Materialized view creation			
36	Packages: Structure and Specification	Package structure and components	Package creation in PL/SQL			
37	Procedures: Syntax and Usage	Procedures for modularizing code	Procedure creation and usage			
38	Triggers: Types and Syntax	Understand trigger functionality and implementation	Trigger programming			
39	KARGIL COMPANY SPECIFIC EXPERIMENT 8 [STORED PROCEDURES IN SQL]	Develop procedures in PL/SQL	PL/SQL procedure programming			
40		Design a stored procedure which take gender as argument, and returns total employee count based on gender passed	PL/SQL procedure programming			
41	PL/SQL Functions	Understand the creation and use of user-defined functions in PL/SQL	Function Creation, Parameter Passing, Return Values			
42		Apply PL/SQL functions for business logic implementation	Function Logic, Conditional Processing, Database Integration			
43		Apply PL/SQL functions for data processing and analysis	Function Creation, SQL Integration, Data Manipulation			
44	Triggers: Types and Applications	Explore types of triggers and their practical use cases	Trigger programming			

45	Packages: Advantages and Developing Modular Programs	Modularize database operations using packages	Modular programming in PL/SQL			
46	Transaction Management: Properties and Serializability	Understand ACID properties and the importance of serializability	Transaction processing fundamentals			
46	KARGIL COMPANY SPECIFIC EXPERIMENT 9 [PACKAGES IN SQL]	Create and implement packages in PL/SQL	PL/SQL package development			
47	Concurrency Control: Locking Techniques	Locking mechanisms for concurrent database access	Concurrency management			
48	Concurrency Control: Timestamping Methods and Recovery Techniques: Need for Recovery	Understand the use of timestamps for managing concurrent transactions , Importance of recovery mechanisms in DBMS and handling failures	Timestamp-based concurrency, Transaction Recovery & Logs			
49	Database Recovery: Types of Errors	Understand various types of database errors and their impact on system recovery	Transaction Failure, System Crash, Logical Errors			
50	Quiz 2	Quiz on PL/SQL	Assessment			
51	KARGIL COMPANY SPECIFIC EXPERIMENT 9 [TRIGGERS IN SQL]	Develop triggers in PL/SQL	Trigger implementation			
52		Design a trigger to implement the automated functionality of Primary Key	Primary key creation with triggers			
53		Implement scalar and vector subqueries to retrieve, filter, and analyze data without using JOIN operations.	Sub-query, Scaler & Vector sub-queries			

54	KARGIL COMPANY SPECIFIC EXPERIMENT 10 [SUB-QUERIES IN SQL]	Exploring two very important types of SQ: Cor-related & Self-contained	Nested queries, Query referencing, Aggregation using Co-related SQ			
55		Understanding unary & binary operators with co-related sub-queries	IN, ANY, ALL with co-related sub-queries			
56	Introduction to SQL Window Aggregate Functions	Understanding SQL window aggregate functions: Interview importantn	SUM(), AVG(), MIN(), MAX(), COUNT(), OVER()			
57	Introduction to Window Analytical Functions	Understanding SQL window analytical functions: Interview important	ROW_NUMBER(), RANK(), DENSE_RANK(), Ntile()			
58	Introduction to Database Indexing: Optimization	Optimizing database queries using DB indexes	Clustered & Non-Clustered Index			
59	Introduction to No SQL (Mongo DB)	Introduction to No SQL: Collections, Documents, Commands	Collections, Documents, JSON, BSON			
60	No SQL Mongo DB CRUD Operations	Introduction to No SQL: Collections, Documents, Commands	Collections, Documents, JSON, BSON			